Heterogeneity in RTA Trade Effects

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Abstract

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1 Introduction

A government weighing a trade agreement asks a question about itself: what will this agreement do to our trade? The empirical literature answers a different question: what have such agreements done, on average, across all pairs of countries that ever signed one. The two questions have the same answer only if agreements affect all pairs alike. If they do not, the average describes the treaty stock rather than any member of it, and the averaging itself does damage beyond answering the wrong question: it pools entities that appear related but differ qualitatively — in what they trade, in how their economies interlock, in what the agreement can reach — and smooths over the distinct causal stories that generated the data (Heckman, 2001). What the questioner wanted was precisely the contingency the pooling removes. Whether this substitution of one question for another is harmless is an empirical matter. This paper argues that in the flagship empirical model of trade it is not, and that its cost has been systematically underestimated because the tools that should reveal it are structurally unable to.

There are good reasons to expect the substitution to matter. An agreement changes the terms under which particular exporters in one economy meet particular buyers in another, and what follows depends on the complementarity of the two economies, the capacity behind their existing trade, and the quality of the match the agreement deepens — determinants no dataset records and no regressor represents. When relevant determinants go unobserved, the influence of the observed ones does not survive untouched: the response to a measured policy is mediated by, and contingent on, the unmeasured conditions under which it operates. Heterogeneity in treatment effects is the statistical name for this contingency — the form that dependence on the unmeasured takes when viewed from inside a regression, represented as variation in the effect precisely because it cannot be represented as a covariate (Pesaran and Smith, 1995). Homogeneity, by contrast, is the assertion that none of these conditions matter: that the same treaty raises trade between neighboring industrial economies and between distant, barely connected ones by an identical proportion. Framed this way, the common-effect specification is not a neutral default awaiting evidence against it. It is the strong assumption, and the burden it has quietly carried in applied work deserves examination rather than presumption.

The substitution would still be tolerable if the estimated coefficient were consumed as a number — reported, compared, filed. It is not. It is consumed as an input, and the chain it enters is nonlinear at every link. The agreement coefficient enters trade-cost functions exponentially, and through them the counterfactual general-equilibrium systems in which the gravity literature conducts its policy analysis, where prices, incomes, and multilateral resistances adjust jointly. Nonlinear transformations do not commute with averaging (Jensen, 1906): the outcome implied by the average effect is not the average of the outcomes implied by the effects, and under convexity, dispersion is not noise surrounding the answer but part of the answer, shifting levels rather than merely widening bands. Nor does the chain end at the model's output. The users of these counterfactuals do not care about a trade elasticity, or even a trade response, for their own sake; they care about what the response does to the things the analysis is meant to inform — revenues, adjustment burdens, the position of an economy vis-à-vis its partners — and that second mapping, from model output to consequence, is nonlinear once more. Two nested nonlinearities amplify whatever the point estimate conceals. Small differences in the estimate, and modest amounts of hidden dispersion around it, compound into large differences in the quantities the exercise exists to compute. Put plainly, a point estimate fed into a nonlinear model is an implicit distributional assumption — it asserts that the distribution of effects is degenerate at a point — and if the true distribution is dispersed, skewed, or tilted across the very pairs the counterfactual concerns, the model returns confident answers about a world that does not exist, with the confidence inherited from an assumption nobody stated.

Nor does the difficulty begin only where the estimate is used; it begins where the estimate is made. "The average effect" reads as a single object, but estimators do not compute unweighted means over pairs. Every fixed-effects structure weights the pairs by the variation left after its absorptions, and different structures — absorbing time, absorbing pair, absorbing country-by-time — leave different variation and therefore weight different pairs. Under homogeneity this is immaterial: all weightings of a constant agree, and the well-documented sensitivity of gravity estimates to specification can only be read as fragility, an embarrassment to be managed. Under heterogeneity the same sensitivity is a prediction. If effects vary systematically along dimensions the weights track, then correctly specified, individually defensible estimators must disagree, and the pattern of their disagreement is information about the shape of what is being averaged. Misread through the common-effect lens, that information becomes something worse than wasted: it becomes an invitation to choose among specifications by their conclusions, with each choice defensible and none interpretable.

The natural escape is to stop averaging: estimate the effect pair by pair and study the collection. But this route fails in a way that is more instructive than the failure of the average, because it fails while passing every test. The instruments used to police unit-level estimates — replication across subsamples, stability across time, the reliability of cross-sectional rankings — all measure persistence. They ask whether an estimate returns when we look again. Yet the dominant distortions in pair-level trade data are themselves persistent characteristics of pairs: biases rooted in volatility and drift do not wash out on a second look; they reappear, and a diagnostic that certifies whatever persists will certify them with the same authority it would certify the truth. Reliable and wrong is a possible state of an estimate, and it is a state the standard toolkit cannot detect from the inside. Confidence, if it is to be had, must be earned differently: not by observing that estimates replicate, but by identifying the mechanisms that would make artifacts replicate, and removing them.

Assembled, the exposures compound. Whoever relies on an average agreement effect is exposed three times over: the average answers a question about the treaty stock rather than the question asked; the nonlinear chain it feeds amplifies whatever it conceals; and the diagnostics that should sound the alarm are structurally silent. None of this is visible in the point estimate or its standard error, which may be small, stable, and beside the point. What would resolve all three exposures is the same object: the distribution of agreement effects across pairs — its center, its spread, its skew, and its variation across the kinds of pairs that policy distinguishes. The resolution is more than formal. Decisions made under uncertainty are rarely symmetric in their stakes: the quantities a decision-maker actually needs are the floor, the probability of being made worse off, and the realistic upside — and these are properties of a distribution's shape that no mean, however precisely estimated, contains. With the distribution in hand, the question a government asks has an answer in the form it was posed; the nonlinear model can be fed what it actually requires; and the disagreement among specifications becomes decomposable rather than embarrassing.

There is, moreover, a reason to expect this distribution to have structure rather than being an amorphous cloud of noise. Economic magnitudes evolve by compounding: growth arrives as relative change applied to an existing state, so that trade between two economies at any moment is the residue of multiplied increments rather than summed ones (Gibrat, 1931; Sutton, 1997; Jones, 2016; Beare and Toda, 2022). A policy acting on such a process plausibly acts the same way — scaling what is there rather than adding to it — and multiplicative action by unmeasured conditions carries testable signatures: lognormal outcomes, asymmetry in effects, effects graded in the size of what they act upon, and selection of agreements on the very conditions that shape them (Mitzenmacher, 2004). These signatures are the bridge between the motivation above and the evidence below: they are what the data would display if the contingency argued for in this introduction is real, and what they should not jointly display if it is not.

This paper estimates the distribution of agreement effects for the flagship empirical model of trade, and makes four contributions along the way. First, we derive the failure of the diagnostics formally: we show why naive pair-level estimation produces estimates that are simultaneously badly biased and nearly perfectly reliable, and we state the conditions under which no persistence-based diagnostic can distinguish artifact from effect. Second, we construct the corrected distribution, removing the identified biases in sequence and the remaining estimation noise by deconvolution; where the data are genuinely silent — on the boundary between transitory effect variation and treatment-coincident volatility — we report a set rather than an assumption-dependent point, which is the same honesty applied at the second moment that motivates the paper at the first. Third, we resolve the disagreement among specifications constructively, showing that the estimates standard structures produce are differently weighted averages of the one distribution we recover, so that the specification spread is explained rather than adjudicated. Fourth, we propagate the distribution through general equilibrium and show that its shape — not merely its mean or its variance — governs the decision-relevant answers: the floor, the downside probability, and the upside; imposing a convenient shape in its place manufactures conclusions, including apparent risks of loss, that the data do not contain.

Section 2 develops the mechanism and states the four predictions it commits to. Section 3 describes the data and the construction of the estimating population. Section 4 presents the naive estimator and its diagnostics, derives the reliability results as formal propositions, and sets out the corrections. Section 5 reports the corrected distribution, its variation across pair types, the reconciliation of the specification spread, and the identification boundary. Section **??** propagates the distribution through general equilibrium. Section **??** concludes and reports which of the four predictions survived. Proofs are collected in Appendix B.

2 Theoretical Framework

2.1 The Economic Primitive

Structural gravity supplies the accounting within which bilateral trade is organized: an exporter's supply capacity, an importer's expenditure, and the trade costs between them measured relative to the costs each faces with every other partner, all disciplined by general-equilibrium consistency (Anderson and Van Wincoop, 2003; Head and Mayer, 2014). That apparatus fixes the size and distance terms and, in the fixed-effects form now standard, absorbs whatever varies by exporter-year and by importer-year. What it does not fix is the pair. Two dyads matched on exporter capacity, importer expenditure and distance routinely trade at volumes that differ by more than the residual variation the model treats as incidental, and the systematic part of that difference is the object of this section. The question is not whether such variation exists but what kind of object generates it, because the answer determines the shape of what a policy acting on such pairs can produce.

Trade between two economies requires that a series of conditions hold at once. What one economy produces must be what the other can use, at the stage of processing at which it is offered; contracts written under one legal order must be enforceable, in practice and not only on paper, under the other; the pair must sit on shipping, payment and correspondent-banking networks that already carry traffic between them; and the administrative apparatus of customs classification, standards recognition and documentation must be in place to a depth sufficient for goods to move without prohibitive delay. These conditions are specific to the pair, are individually unobserved, and no dataset records them. Decisively, they do not substitute for one another (Kremer, 1993). A shipment occurs because every link in the chain holds; an unusually complementary production structure does not compensate for an unenforceable contract, and a well-functioning payments corridor does not compensate for goods the importer cannot use. Conditions of this kind enter the outcome multiplicatively rather than additively, and the compounding is not a modeling convenience but a restatement of what it means for a chain to hold. Multiplicativity is itself a consequence of heterogeneity under complementarity: when the value of an activity depends on which others are present, the combinations available expand as a function of those already realized, and growth that builds on the existing stock in this way is multiplicative by construction.

This structure is neither peculiar to trade nor unfamiliar. It is the production structure of the weak-link literature, in which output under complementarity is governed by the least functional element rather than by the average of the elements, so that a defect at any point in a chain reduces the value of competence everywhere else (Kremer, 1993; Jones, 2011). The same logic organizes the older capital-theoretic treatment of complementarity, in which the productivity of an addition to the capital stock depends on the composition of the stock it joins rather than on its own quantity (von Böhm-Bawerk, 1921; von Hayek, 1941; Lachmann, 1956). More broadly, economic processes are characterized by mechanisms that reinforce what is already in place: network effects and increasing returns that compound over time (Pietronero et al., 2001; Gabaix, 2009); agglomeration (Krugman, 1991); and learning-by-doing, in which capability is a function of accumulated activity (Arrow, 1962; Romer, 1986). In each case the marginal contribution of a factor is conditioned on the configuration of the others rather than being fixed. Taking the bilateral pair as the unit at which such configurations are realized, the residual variation left by structural gravity is what a multiplicative interaction of unobserved pair conditions would look like when viewed through a specification that has no place to put them.

2.2 From Primitive to Distribution

If bilateral outcomes are the product of many pair-specific factors that vary independently enough not to move as one, the logarithm of the outcome is a sum of many varying components, and the distribution of outcomes across pairs belongs to the lognormal family — Gibrat's law of proportionate effect applied to the dyad rather than the firm (Gibrat, 1931; Sutton, 1997). This is the mechanism's first prediction: (i) bilateral trade flows are distributed lognormally across pairs, with the right-tail weight that family implies. The prediction is genuine, in that a mechanism generating additively separable conditions would not produce it, but its discriminating power is limited and it would be dishonest to present it otherwise. Lognormal outcomes are overdetermined: random proportionate growth, the aggregation of heterogeneous sub-units, and several other generative stories produce the same signature, and distinguishing among them from the shape of a single cross-section is generally not possible (Mitzenmacher, 2004). Confirming (i) therefore removes rival mechanisms that predict something else while leaving standing every rival that predicts the same thing, which is why this paper does not rest its case on this signature, and why the mechanism must be made to commit to predictions about the treatment effect and not only about the outcome.

2.3 Non-Separable Treatment Effects

The empirical models used to estimate trade flows are themselves multiplicative. The workhorse specification writes the flow as $y_{ij} = \exp(\mathbf{x}_{ij}'\boldsymbol{\beta} + \epsilon_{ij})$, estimated by Poisson pseudo-maximum likelihood under $\mathbb{E}[e^{\epsilon_{ij}} \mid \mathbf{x}_{ij}] = 1$ (Santos Silva and Tenreyro, 2006). The functional form is multiplicative but the shock is not: ϵ_{ij} enters in parallel to the covariates, so that whatever the pair's unobserved conditions do to its trade, they do independently of the policy variable sitting in $\mathbf{x}_{ij}$. An agreement then shifts every pair's trade by the same proportion, and the pair conditions of Section 2.1 are confined to setting the level from which that common proportion is taken. This is a substantive restriction, not a normalization. It asserts that the conditions determining whether the chain holds have no bearing on what happens when one link in the chain is loosened.

The alternative is to let the covariates interact with the shock, $y_{ij} = \exp(\mathbf{x}_{ij}'(\boldsymbol{\beta} + \boldsymbol{\epsilon}_{ij}))$, under the corresponding normalization $\mathbb{E}[e^{\mathbf{x}_{ij}'\boldsymbol{\epsilon}_{ij}} \mid \mathbf{x}_{ij}] = 1$, which preserves the familiar conditional mean $\mathbb{E}[y_{ij} \mid \mathbf{x}_{ij}] = e^{\mathbf{x}_{ij}'\boldsymbol{\beta}}$ and which we maintain throughout. This form has two readings. Structurally, it is a random-coefficients model in which the effect of each covariate, the agreement among them, varies across bilateral pairs, with the variation governed by the distribution of $\boldsymbol{\epsilon}_{ij}$. Empirically, it generates heteroskedasticity of the kind long documented in trade data (Santos Silva and Tenreyro, 2006): pairs facing higher systematic trade costs are more exposed to idiosyncratic frictions in implementation, and therefore deviate more widely, in absolute terms, from what the model predicts for them. Written in terms of the object of interest, the agreement effect for a pair is

$$\theta_{ij} \;=\; f(A_{ij},\, c_{ij})\,, \tag{1}$$

where A_{ij} denotes the agreement and c_{ij} the pair's unobserved conditions — rather than a scalar θ common to all pairs. Neither argument of f is homogeneous. Agreements enter the empirical literature as a binary indicator, but the legal instruments so indicated range from tariff schedules with little else attached to architectures governing services, procurement, standards and dispute settlement, and treating these as one object imposes a second homogeneity assumption alongside the first; we do not attempt to separate the two sources of dispersion here, and note only that both are suppressed by the same specification and neither is consistent with a scalar θ. The economics of the modulation is the economics of a chain. An agreement does not create trade directly; it relaxes a constraint, lowering a tariff, recognizing a standard, binding a dispute procedure. Whether relaxing that constraint moves anything depends on whether the other links were holding, which is precisely what c_{ij} records. The effect is therefore modulated by the pair's conditions rather than added to them, and three further predictions follow from that modulation.

The first concerns shape. If the effect is the product of the relaxation and the conditions it acts through, the effect distribution inherits the compounding of the conditions and is therefore right-skewed rather than symmetric: (ii) most pairs gain modestly, a minority whose conditions are aligned gain multiplicatively, and the left tail is thin. The asymmetry has a specific source, and it is worth stating plainly because it is what distinguishes this prediction from a generic appeal to skewness. The upside is unbounded, in the sense that the conditions can reinforce one another without limit; the downside is bounded, because an agreement that fails to bind is inert rather than destructive. A pair whose chain was broken before the agreement, and remains broken after it, records no effect — it does not record a loss. There are channels through which an agreement could reduce a particular pair's trade, diversion toward a third partner most obviously, but these operate against the floor rather than mirroring the upside, so the mechanism predicts an effect distribution with a short left tail and a long right one.

The second concerns gradation, and it is the prediction that gives the mechanism purchase on observable data. If an agreement works by relaxing constraints, its effect is largest where the constraints bind hardest. But the constraints a treaty addresses are discrete rather than continuous: a tariff line, a standards recognition, a documentation requirement. Each either still obstructs the pair or has already been surmounted, and a pair trading at volume through established channels has, by revealed behavior, already surmounted them — the fixed costs of compliance are sunk, the customs relationships exist, and the trade would not be occurring otherwise. Relaxing what has already been worked around changes little. The prediction is therefore that (iii) effects are graded in pair size and prior integration, but not smoothly: they are large where the treaty's constraints still bind and fall to a common floor once they do not. What the mechanism forbids is continued decline among pairs that have already exhausted the margin, since beyond exhaustion there is nothing left to grade. This is sharper than (i) and (ii), because it names both a direction and a shape against variables the data contain, and it is falsified not only by a positive slope but by a smoothly monotone negative one.

The third concerns who signs. The conditions in c_{ij} are not observed by the econometrician, but they are observed, imperfectly and in the aggregate, by the governments and the exporters who lobby them. Agreements are negotiated where there is something to negotiate over, which is to say where enough of the chain already holds that the parties can foresee gains from relaxing what remains. The prediction is that (iv) agreements are selected on the same unobserved conditions that modulate their effects: treated pairs are drawn from the stable end of the condition distribution rather than at random from it. This is assortative matching of the kind that complementarity produces generally, where partners of similar reliability are matched together in equilibrium precisely because reliability is complementary (Kremer, 1993), applied to governments choosing whom to negotiate with. Two consequences follow, and they should be stated together because they pull in different directions. Any estimate recovered from treated pairs is an effect on self-selected matches and does not license extrapolation to a pair that has not signed, which limits the external reach of what the sections below can report. But the same selection is a testable implication rather than only a limitation: it predicts that the treated population is more stable in its pre-agreement behavior than the untreated, and that prediction can be checked without observing c_{ij}.

Predictions (iii) and (iv) may appear to pull against one another, the first locating the largest effects among small and thinly integrated pairs, the second locating agreements among pairs whose conditions already hold. They operate, however, on disjoint parts of c_{ij}. Selection runs on the links a treaty does not address: contractual enforceability, payment and correspondent-banking infrastructure, the administrative capacity of the counterparty — the conditions that make gains foreseeable and therefore worth the cost of negotiating. Gradation runs on the links a treaty does address: tariffs, administrative barriers at the border, standards recognition — the constraints still binding that an agreement can relax. A pair may have a functioning non-policy chain, and so be a plausible negotiating partner, while its policy-addressable constraints remain fully binding and its realized volume correspondingly low. That combination describes a small, thinly integrated pair with a competent counterparty, and it is not unusual. The two predictions are therefore compatible, and their conjunction is sharper than either alone: the mechanism predicts that the treated population is drawn from the stable end of the condition distribution and not from the large end, and stability and size are separable in the data.

The distinction between the non-separable form and the additive alternative, in which the shock enters in parallel to the covariates rather than multiplying them, is consequential rather than conventional: only the non-separable form generates dispersion through the interaction of the shock with treatment, and only it implies that the dispersion of the outcome varies with treatment. That implication is testable directly, on pairs observed before and after adoption, and the test bears the weight of everything that follows: if the additive form survives it, the modulation in Equation (1) is a redescription rather than a mechanism, and the corrections built upon it are unmotivated. We defer the argument for the non-separable structure, and the test that distinguishes it from the additive alternative, to Section 4.1, where it underpins the analysis of how the adjustment interacts with the fixed-effects structure.

2.4 The Evidentiary Standard

Each of these four predictions is weak taken alone, and we ask the reader to hold the argument to the standard the mechanism actually meets rather than to one it does not. Lognormal outcomes are overdetermined, as Section 2.2 conceded. Right-skewed effects are consistent with any story in which gains are bounded below and unbounded above. A size gradient is consistent with diminishing returns of a wholly different origin, or with measurement error correlated with volume. Selection on unobservables is consistent with almost any account of why governments negotiate. What the mechanism claims is not that any one of these is decisive but that their conjunction is jointly predicted: lognormal outcomes *and* right-skewed effects *and* effects graded in size and prior integration *and* selection on the conditions that do the modulating. Each rival that explains one of these comfortably explains at least one of the others badly or not at all, and to our knowledge no single alternative currently on the table predicts all four together. This is abduction, not proof: the conjunction is evidence for the mechanism in the sense that the mechanism would render it unsurprising and its rivals would not, and a mechanism we have not considered may do as well or better. Sections 4 and 5 test each prediction in turn — (i) in the distributional diagnostics, (ii) and (iii) in the corrected effect distribution and its variation across pair types, and (iv) in the composition of the treated population — and Section **??** reports which survived.

We state the outcome in advance, because the conjunction is only as strong as its weakest leg and the reader should know from the outset which leg that is. Lognormal-type outcome tails are not tested here. The prediction concerns the marginal distribution of flows, and this paper's design bears on it only through the distributional assumption the propositions maintain; that assumption is examined directly in Section 4, and it is rejected. Effects graded in size and prior integration are confirmed, and in the specific form this mechanism requires — a steep decline across the less integrated pairs followed by a common floor, not a smooth monotone descent, which would count against the mechanism rather than for it. Selection on match quality is supported: the pairs that adopt are systematically unlike the pairs that do not, in ways that persist and that the correction layer detects without fully purging. Right-skewed effects are neither confirmed nor refuted. The shape of the effect distribution is not identified at the signal share these data achieve, and we report that as a limit of the measurement rather than as evidence against the prediction. Two legs carry weight, one is under-identified and one is untested; we take the conjunction to be informative on that basis and not on a stronger one.

3 Data and the Estimating Population

Section 2.4 committed the mechanism to four predictions and assigned each to a later section. Those assignments are obligations on the data as much as on the method. A prediction about the shape of outcomes, about gradation in size and prior integration, and about the composition of the treated population can be tested only against a population whose construction is fixed in advance and stated in full, since a construction chosen after the estimates are seen can accommodate any of them. This section fixes it. We describe the trade panel and the two configurations in which it is used, the agreement indicator and how adoption is dated from it, the rules that select the pairs on which pair-level effects are computed, and the populations from which the counterfactual is built. Nothing here reports an estimate.

3.1 Trade Flows

Bilateral trade comes from the International Trade and Production Database for Estimation, Release 2 (Borchert et al., 2022), aggregated to total goods trade for each ordered exporter–importer pair and year, in millions of current United States dollars. The panel runs from 1988 to 2019, which is the window over which the release's goods sectors are jointly available; its services industries begin only in 2000 and are outside the panel by coverage rather than by choice. After the restrictions described next, the estimation panel contains 794,720 observations across 46,803 ordered pairs, of which the terminal cross-section contributes 31,650. These counts and the year bounds are frozen: every script verifies them against the source file before computing anything and halts if they do not reproduce.

Two configurations of this panel are used, and we state both here. Estimation of the agreement effect uses positive international flows: observations in which exporter and importer coincide are excluded, and observations recording zero trade are excluded with them. The first follows from what is being estimated — an agreement acts on trade between two countries, and a country's trade with itself carries no agreement — and the second from the pair-level construction, which takes logarithms of realized flows and their counterfactual counterparts. The general-equilibrium analysis of Section **??** re-introduces domestic flows, because its accounting requires them: an expenditure system in which each economy's outlay must exhaust across all sources cannot be closed on international flows alone. The two are not competing samples but the input requirements of two different calculations.

One property of the release deserves recording where the exclusions are stated rather than buried. Missing international flows in Release 2 are set to zero on the reasoning that reported trade is comprehensive and the bilateral matrix is genuinely sparse. A zero here is therefore an observed absence of recorded trade rather than a measured value of nothing, and dropping zeros drops both. We take no position on the imputation and note only that it makes the boundary between a pair that does not trade and a pair whose trade is unrecorded less sharp than the data's appearance suggests.

The outcome is the gross value of goods shipped from one member of the pair to the other. This is the right object rather than merely the available one. The treaty is a bilateral instrument: it is signed by two governments, it alters the terms on which exporters in one meet buyers in the other, and the unit at which its provisions bind is the pair. An outcome on any other unit would either aggregate across pairs the treaty treats differently or decompose the pair into components the treaty does not address. Value-added measures reallocate part of a bilateral relationship's response to the third countries supplying its inputs, which is informative about where production sits but not about what the agreement did to the relationship it governs; netting flows within a pair discards the directional asymmetry the provisions themselves distinguish. Gross directed flows keep estimand and instrument on one unit.

3.2 Agreements

Agreement status is the binary indicator of Egger and Larch (Egger and Larch, 2008), recording for each pair-year whether a reciprocal agreement is in force. Pairs are classified from the time path of this indicator alone. A pair is a *switcher* if the indicator makes exactly one transition from zero to one and none from one to zero; its adoption year is the first year the indicator equals one. Pairs whose indicator equals one in every year are always-treated: they carry no within-pair variation and cannot contribute to a within-pair design, whatever their weight in world trade. Pairs recording more than one transition are excluded, because adoption is not well defined for them and a pair-level effect computed across their regimes would average over the very heterogeneity the paper measures. Together these exclusions are large: 6,339 of 46,803 pairs are single switchers, and the 40,464 removed are overwhelmingly never-treated.

The indicator is binary, and Section 2.3 has already observed that a single indicator bundles legal instruments ranging from bare tariff schedules to architectures governing services, procurement, standards and dispute settlement; the construction described here does not separate variation originating in that bundling from variation originating in pair conditions, and we make no claim about their relative contribution.

3.3 The Estimating Population

The design is within-pair: each pair's effect is identified from the comparison of its own realized trade after adoption with what the fitted model implies for it in the absence of the agreement, so the pair is its own control and no cross-pair comparison enters the pair-level estimate. Every selection rule follows from that design, and each is stated with the reason it exists rather than as a threshold applied by convention. Table 1 reports the pairs surviving each rule in the order applied.

The rules are: (a) *switchers only*, since the within-pair comparison requires observation on both sides of adoption, and always-treated and multi-switcher pairs are excluded by Section 3.2. (b) *adoption within 1991–2016*: a pair adopting in the panel's first or last years cannot satisfy the window requirements below, and the range makes the boundary explicit rather than implicit; it removes 753 pairs. (c) *a symmetric one-year exclusion band around adoption.* The pre-period ends at adoption − 2 and the post-period begins at adoption + 2; the adoption year and both flanking years are assigned to neither side. Agreements are announced before they enter force and implemented in stages after, so trade in the immediate neighbourhood of the switch belongs cleanly to neither regime. The band is symmetric by construction, and the same band governs the comparison populations of Section 3.4, so that treated and untreated pairs are never compared across differently drawn windows.

Two further rules concern estimability and window length. (d) *a usable counterfactual*: a pair enters only if the imputation model of Section 4 yields a prediction for its post-adoption cells, which requires that its own fixed effect be identified from untreated data. This removes 576 pairs; because identification binds at the pair fixed effect rather than cell by cell, every retained pair has a counterfactual for all of its post-adoption years, and no retained pair is estimated on a truncated window. (e) *at least three pre- and three post-adoption years of positive trade*, outside the exclusion band, removing 353 and 475 pairs respectively. A pair-level effect estimated on one or two years is dominated by whichever shocks fall in them, and the minimum buys enough within-pair observations for the effect and its sampling error to be separately estimable. Robustness across the threshold grid is reported in Appendix A [X-REF: threshold grid].

Rule (e) carries a consequence that should be stated rather than left to be discovered. A pair recording no positive trade in any pre-adoption year has no pre-period and is removed by the rule mechanically; there is no separate extensive-margin exclusion because none is needed. The effect this paper estimates is therefore an intensive-margin effect — what an agreement does to trade between countries already trading — and it is silent on the creation of trading relationships where none existed. That is a real restriction on the estimand's reach, since the excluded pairs are precisely those for which an agreement might matter most in proportional terms.

Applying these rules yields the canonical population of 4,182 pairs, with adoption years from 1992 to 2015. The population is economically substantial: its members account for 22.1 percent of world international goods trade in 2015. Its composition, however, deserves comment. The largest members by pre-adoption trade are the Korea–United States agreement of 2012, the Australia–Japan and Australia–China agreements of 2015, the China–Hong Kong arrangement of 2003, and the Japanese agreements with Thailand, Indonesia and Singapore; the ten largest directed pairs account for 34.3 percent of the population's pre-adoption trade. Conspicuously absent are the North American and European agreements a reader might expect. They are absent by construction rather than by selection: Canada and the United States were already in force under their 1989 agreement when the panel opens, and intra-European pairs are in force throughout, so all are always-treated and carry no within-pair variation to identify from. The population is thus weighted toward the Asia-Pacific agreements of the 2000s and 2010s, and results should be read as speaking to agreements of that kind.

With the population fixed, the object of estimation can be stated exactly. It is the distribution across adopting pairs of the pair-level multiplicative effect of a new agreement on bilateral trade, for pairs with pre- and post-adoption trading histories.

3.4 The Comparison Populations

Two further populations are constructed here and used later. The first is the never-treated pairs: those whose indicator is zero throughout, to which pseudo-adoption years are assigned under the same band and minimum-length rules that govern the switchers. Because nothing happens to these pairs at their assigned dates, whatever a pair-level effect returns for them is by construction not an agreement effect, and the population therefore serves as a reservoir against which the machinery can be calibrated. The second is the switchers' own pre-adoption periods, which record how each pair behaves in the absence of the treatment it later receives, and so allow the pairs to calibrate themselves rather than each other.

These two populations have a further role that belongs here rather than later, because it determines which observations are used at all. Their union — every year of every never-treated pair, together with every pre-band year of every switcher — is the sample on which the counterfactual is fitted. No cell in which an agreement is in force enters that fit. The consequence is that a switcher's post-adoption counterfactual is a prediction for years the fitting sample never saw, rather than a fitted value for years it did; what that buys, and what it costs, is the business of Section 4. We record here only the sample definition, since it is a property of the data construction and not of the estimator applied to it.

4 The Anatomy of the Artifact

The counterfactual of Section 3.4 makes a pair-level estimate available without further apparatus. For each switching pair, take the untreated-only PPML fit, impute the trade the pair would have carried in each of its post-adoption years, and average the log gap between realized and imputed trade over those years. Call this Definition A. Across the canonical population it returns a mean of -0.2837, a standard deviation of 1.6630, and

Table 1: Construction of the estimating population

	Rule	Pairs	Removed
	All ordered pairs, international, positive trade	46,803	—
(a)	Single switchers (one $0 \to 1$ transition)	6,339	40,464
(b)	Adoption year in $[1991, 2016]$	5,586	753
(d)	Usable counterfactual from untreated data	5,010	576
(e1)	≥ 3 pre years, outside the band	4,657	353
(e2)	≥ 3 post years, outside the band	4,182	475
	Canonical population	**4,182**	

Notes. Rules are applied in the order shown. Rule (c), the symmetric one-year exclusion band, is not a separate row: it defines the pre- and post-periods over which (e1) and (e2) count years, so its effect is embedded in those two rows. Pre years are those before adoption $-$ 1 and post years those after adoption $+$ 1, both requiring positive trade. Panel totals are frozen and verified at load. Source:.

a split-half correlation across disjoint halves of each pair's post window of 0.9243, on the 4,120 pairs with at least two post-adoption years in each half, or 0.9607 once corrected for split length. The estimator is thus sign-wrong against everything the aggregate literature reports, enormously dispersed, and—by the one diagnostic that pair-level estimation makes available—close to perfectly reliable.

The never-treated pairs of Section 3.4 carry pseudo-adoption dates assigned under the same band and minimum-length rules, and nothing whatever happens to them at those dates. Run Definition A on them unchanged. On the 15,683 never-treated pairs that qualify for the same split, the mean "effect" is -0.6821, with a standard deviation of 1.0814 and a split-half correlation of 0.7463, or 0.8547 corrected for split length. A population in which the effect is zero by construction thus reproduces roughly four-fifths of the treated estimator's reliability—0.7463 against 0.9243, computed under the same split rule in the same run —and rather more than twice its mean in magnitude.

The reason is not subtle once stated, and it is a property of the diagnostic rather than of this application. Split-half reliability asks whether a pair's estimate in one half of its post window predicts its estimate in the other. Any quantity that is a persistent trait of the pair will answer yes. The artifacts at work here—each pair's own volatility, and each pair's own trend relative to its fitted level—are exactly such traits: they are properties of the relationship, not of the years sampled from it, so they replicate across splits as faithfully as a real effect would. Proposition 1(c) gives the exact result under the stated assumptions: the split-half correlation of a zero-effect estimator equals the share of its variance carried by persistent bias, $\mathrm{Var}(b)/\big(\mathrm{Var}(b) + \mathbb{E}[\sigma^2]/T_h\big)$. Corollary 1 states the consequence without reference to any particular artifact: *any* dispersed, persistent per-pair bias passes. High reliability certifies persistence. It does not certify validity, and no amount of it can, because the two are different questions and only one of them is being asked.

It is worth being precise about what the placebo population establishes and what it does not. It establishes that the machinery, applied to pairs where the answer is known to be zero, returns numbers that are large, dispersed, and stable across resplits, and that the diagnostics conventionally used to adjudicate such estimates do not distinguish that case from a real one. It does not establish that the treated estimates are artifact; the treated pairs received agreements and the never-treated pairs did not, and nothing in the comparison speaks to what the agreements did. What the comparison does is set the standard the treated estimate has to clear, which is not zero but whatever the same machinery returns when there is nothing to find. The rest of this section is the accounting that clears it, layer by layer, and the accounting is worth doing precisely because the diagnostics that would ordinarily settle the matter have been shown not to.

What follows takes the naive estimate apart into three layers. Each has a mechanism with a derivation, a correction, and a population in which effects are absent by construction and the mechanism is therefore visible on its own. That last requirement is the justification standard we hold the corrections to throughout: a subtraction is licensed when there exists a population where the quantity subtracted is all there is to find. The contrast with the conventional instrument is worth making once. Fixed effects remove identifying dimensions before estimation, and what they remove is unavailable for inspection afterwards; the corrections below remove what the estimator is demonstrated to report when the effect is zero, and the demonstration is a table one can read.

The first layer is aggregation. Definition A averages log ratios, and the logarithm is concave, so a pair whose realized trade is volatile around its counterfactual is penalized for the volatility itself. Under the assumptions of the appendix and with zero treatment effect, $\mathbb{E}[\theta^A_{ij} \mid ij] = -\sigma^2_{ij}/2$ exactly: the estimator returns a negative number whose magnitude is half the pair's own log-variance and has nothing to do with any agreement. The remedy is to aggregate before taking the logarithm rather than after—the log of the ratio of sums over the post window, Definition B—which replaces a pair-level penalty of order σ^2_{ij} with one of order $\mathrm{Var}(R_{ij}) = \mathrm{Var}(\eta_{ij}) \sum_t w_t^2$, smaller by roughly the number of post years.

The move is decisive for the mean and only for the mean. Across the canonical population the average effect goes from -0.2837 under A to 0.0758 under B: the sign flip happens here, at an arithmetic layer, before any correction that uses information about untreated pairs. The zero-effect population moves in the same direction and by more. Measured on the same 17,200 never-treated pairs by the same producer, its mean rises from -0.6880 under A to -0.2072 under B. It does not, however, reach zero. Against the validity threshold of 0.05 carried by the pipeline's gate library, the uncorrected placebo mean under Definition B sits at roughly four times the threshold. That residual is systematic rather than incidental—it is a property of the population and the window, not of any resampling—and the second layer is what it is made of.

The second layer is within-pair drift. Trade relationships do not hold station around their fitted level; they fade or strengthen relative to it, and a pair fixed effect centers the level without touching the trend. Any comparison of a late window to an early one therefore reads a fading relationship as a negative effect. Proposition 2(a) makes the size of the misreading explicit: for a never-treated pair with a post window equal to the last T_{post} years of its span, the post-window mean of centered time is $T_{\text{pre}}/2$, so $\mathbb{E}[\theta^A_{ij}] = -\sigma^2_{ij}/2 + \delta_{ij}T_{\text{pre}}/2$. Two signatures follow and are jointly restrictive. The bias depends on window geometry alone and never on calendar time, so it cannot be a macroeconomic episode in disguise. And where drift correlates with pair size, the bias is graded in size—which is what makes a size-cell correction the right shape of instrument rather than an arbitrary one.

The correction estimates, within size cells, the mean effect the machinery returns on never-treated pairs, and subtracts it. Two features make this more than a relabeling. It is estimated on held-out placebo splits and applied out of sample, so no pair contributes to its own correction. And it is validated on the held-out half, where the corrected placebo effect should be zero if the subtraction removed an artifact and nothing else. It is: across 8,545 validation pairs the mean corrected placebo effect is 0.0188 with a standard error of 0.0087, inside the 0.10 decile bound of the pipeline's three-state gate and inside the stricter 0.05 bound as well. One of the ten size deciles does not clear the first bound. The third decile returns -0.1013 on 314 validation pairs, with a standard error of 0.0591, which registers as PARTIAL under the three-state rule—beyond 0.10, well inside 0.20, and 1.7 standard errors from zero. We report it as such rather than absorbing it into an average that would hide it. Applying the correction moves the mean from 0.0758 under B to 0.2473 under D, with a standard error of 0.0241.

The unit of the correction is worth dwelling on, because it is also the source of the layer's limitation. Proposition 2(c) is explicit that subtracting a size-cell mean removes $\mathbb{E}[\delta_{ij} \mid s]T_{\text{pre}}/2$ and leaves the within-cell dispersion of drift untouched, so that $\operatorname{Var}(\theta^D \mid s)$ still contains $(T_{\text{pre}}/2)^2 \operatorname{Var}(\delta_{ij} \mid s)$ alongside the true effect variance and the sampling term. A finer partition would remove more of it. The binding constraint is that the correction must be estimated on held-out placebo pairs and validated on them, and the validation folds are already thin at the current resolution: the smallest size decile contributes 19 validation pairs and the second 94, against 1,365 in the eighth. Refining the partition would buy a smaller residual at the cost of a correction no held-out population is large enough to certify, which is the trade the current design declines to make. The consequence is carried forward: whatever drift heterogeneity survives within cells is still in the dispersion, and the next layer has to account for it rather than assume it away.

The two corrections between them move the mean by roughly half a log point. They do almost nothing to the spread. The standard deviation is 1.6630 under A, 1.5799 under

B, and 1.5614 under D —a decline of about six percent across a path that reversed the sign of the mean. This is not a failure of the corrections. It is a statement about what they were built to remove: both are corrections to a location, the first to a pair's own expected penalty and the second to a size cell's average drift, and neither has any purchase on variation across pairs within a cell. The dispersion that remains is a different object and has to be accounted for separately.

4.1 The null accounting

That accounting proceeds by asking how much of the observed dispersion a zero-effect population reproduces, and the answer depends on which zero-effect population one is entitled to use. Against a total variance of 2.4380, three admissible nulls have been computed. Sampling noise from finite post windows, augmented by the uncertainty in the estimated counterfactual itself, accounts for 0.2611, or 11 percent; the counterfactual component is a third of that total and cannot be dropped. The in-sample placebo null—the dispersion the machinery generates on never-treated pairs over the same years—accounts for 0.6796, or 28 percent. The out-of-sample drift null, computed under the canonical symmetric-window design so that treated and pseudo-treated pairs are compared across identically drawn windows, accounts for 1.887, or 77 percent.

These three shares do not partition anything and must not be added. The nulls are nested by construction: the first admits sampling variability and the uncertainty of the imputation and nothing else; the second admits those together with whatever the machinery generates on pairs that received no agreement over the same calendar years; the third admits those together with the window-geometry drift of Proposition 2, measured on pseudo-treated pairs whose windows are drawn as the treated pairs' windows are. Each is the previous one plus a further admitted source, which is why they run 11, 28, and 77 percent rather than in any other order, and why the middle one is an interior value rather than an endpoint. Choosing among them is choosing how much of the observed spread one is prepared to concede to a mechanism that produces spread without producing effects, and the propositions do not settle that choice—they establish what is at stake in it.

The first and third of these are the admissible endpoints, and they bracket the standard deviation of true effects at [0.74, 1.48]. We report the interval and no point inside it, and the reason is not caution but Proposition 3. Part (a) identifies the variance of persistent effects as the cross-pair variance of the estimate minus the expected within-pair variance over the post window, which is estimable. Part (b) shows that the composition of that within-pair variance is not: under the stated normality, transitory effect heterogeneity and a treatment-coincident change in volatility enter the likelihood only through their sum, so any two parameterizations with equal composites generate identical data distributions. The split between them is a modeling label, not an estimable quantity, and the two endpoints are what the two admissible labels imply. The interval is therefore the dispersion conclusion of this section, with each arm's assumption attached to it: the upper endpoint concedes only noise, the lower concedes everything an out-of-sample zero-effect population reproduces. One feature of the lower endpoint deserves emphasis, because it is what makes that endpoint conservative rather than merely small. The out-of-sample null is not formed by applying an already-fitted counterfactual to held-out years. The counterfactual is re-estimated from a sample from which the evaluation window has been excluded entirely, so that no cell used to judge the null contributed to the fit that generates it. This matters more than it may appear. A null built the other way—reusing a counterfactual fitted on the very cells it is then evaluated against—reproduces substantially less of the observed dispersion, because the fit has already absorbed part of what the null is meant to measure, and it would place the lower endpoint considerably higher. We report the construction that is harder on the result.

4.2 The certificate

A correction that removes an artifact and a correction that removes the signal look identical in a table of means. The distinguishing evidence has to come from data whose truth is known in advance: the propositions are evaluated numerically, with parameters planted and recovered, and the evaluation is run at the parameter values where the derived approximations are supposed to fail as well as where they are supposed to hold. What that establishes is that the arithmetic underlying each correction is sound and that its stated failure modes occur where the derivation says they will.

The first is available in full. Each proposition is evaluated numerically at the ledgered dispersion, which the zero-effect population identifies directly, since the mean log-ratio there is exactly minus half the mean log-variance. Three of the evaluations plant a parameter and recover it. The Jensen expansion is checked where it is weakest: at the empirical scale the generic second-order form is invalid for Definition A altogether, and for Definition B it overstates the bias magnitude by about thirty percent, the positive third-order term partially offsetting the variance term. The drift formula is checked by planting a per-pair trend and recovering the window-geometry prediction. And the reliability decomposition is over-identified, which permits a check with nothing fitted to it: the cross-pair variance of the estimator on the zero-effect population implies a dispersion of pair volatilities, and substituting that into the split-half formula *predicts* a reliability of 0.754 against the 0.746 observed. The formula reproduces the observed figure to within eight thousandths, with no parameter chosen to make it do so. The prediction is evaluated pair by pair, which the convexity of $1/T$ requires: post windows differ across pairs, the reciprocal of the mean window length is not the mean of the reciprocals, and a plug-in evaluation at the mean over-predicts, returning 0.807 and a gap eight times larger.

The most informative check is of an estimator variant that does not work. A null calibrated on a window shorter than the window over which the estimate is formed over-subtracts by a computable factor $\kappa_w := T_{\text{post}}/T_0 > 1$, biasing the recovered variance by $-(\kappa_w - 1)\mathbb{E}[V^{\text{post}}]/T_{\text{post}}$. The consequence is exact and can be read off. At $\kappa_w = 1$ the subtraction is the noise-only arm and returns its dispersion of 1.475 by construction; at $\kappa_w = 2, 3, 5$ the recovered dispersion falls to 1.384, 1.286 and 1.064; and at $\kappa_w = 9.34$ the entire estimated heterogeneity is absorbed. A mis-specified null window therefore consumes true heterogeneity at a rate the derivation gives in closed form, and a sufficiently mis-specified one consumes all of it while returning a number that looks like an answer. That the failure occurs at the magnitude the derivation predicts is a stronger statement than a variant that succeeds, and it is the reason the null used above is matched to the treated design rather than to convenience.

One further check bears on the premise rather than the arithmetic. The propositions are derived under a distributional assumption — that the log gaps are normal within a pair — and an assumption of that kind is testable on the untreated cells, where the effect is zero by construction. It is tested here against a matched normal control rather than against a theoretical one, because centering and scaling within a pair over a short window deforms the distribution even when the underlying data are exactly normal; the control is drawn pair by pair with identical cell counts and passed through identical code, so any deformation the procedure introduces appears on both sides and cancels. On 27,782 pairs the assumption is rejected: 32.8 percent of pairs reject normality at the five percent level against 5.0 percent for the control. The direction matters more than the rejection. The departure is left-skewed and leptokurtic — skewness -0.476, excess kurtosis 0.784 against the control's -0.269 — while the *upper* tail is thinner than the control's at the ninety-fifth, ninety-ninth and ninety-nine-and-a-half percentiles. What the data carry is not an excess of extreme gains but an excess of collapses: trade falling far below its counterfactual more often than a normal law allows.

The consequence for what precedes it is smaller than the rejection suggests, and worth stating in those terms rather than defended. The reliability prediction above is derived exactly under normality, and it lands within eight thousandths on the same data in which normality is rejected at a third of pairs. A prediction that survives the failure of its own premise is evidence about the mechanism it describes rather than about the premise, which is the reason this section reports the departure alongside the check instead of in place of it. Where the assumption does bind is on the identification boundary rather than on any quantity reported here; Remark C states what the measured departure recovers there, and the section on the boundary of learnability carries it forward.

What this does not establish should be stated with equal clarity. Verifying the arithmetic of a correction is weaker than verifying the pipeline that applies it. The stronger test simulates panels carrying a known effect distribution alongside a known artifact, runs the estimator and both corrections without modification, and asks whether the recovered distribution matches the planted one while the planted artifact is returned as zero; a pipeline that passes the first requirement and fails the second is absorbing signal, and one that passes the second and fails the first is leaving artifact behind. We report the propositions' verification and do not claim that end-to-end test, which we regard as the standard such corrections should eventually be held to rather than one this paper meets.

4.3 Diagnostics that survive

The negative result of this section is narrow: reliability statistics answer a question about persistence and were never answering the question they are conventionally read as answering. The constructive one is that the same machinery, pointed at the right comparison, is informative. Five practices follow, and the first fixes the diagnostic that failed. A reliability statistic is uninterpretable in isolation and interpretable against its own null: report the split-half correlation of the estimator on a population where effects are absent by construction, computed under the identical split rule in the same run, and read the treated figure against it. The contrast is the statistic; either number alone is not.

Second, a pair's own pre-adoption years are a null it calibrates for itself, and pseudo-effects estimated on them measure what the machinery returns from that pair when there is nothing to find. The out-of-sample drift null used above is this instrument at population scale, and its discipline is that the window geometry of the null must match the window geometry of the estimate—the comparison is uninformative, and can be badly misleading, when the two are drawn differently. Third, verification on planted truth is the certificate a pipeline cannot supply about itself from observational data alone. Its value is not confirmation that the estimator works; it is the location of the parameter region where it stops working, which is knowable in simulation and not knowable otherwise.

Fourth, where point identification fails, the honest object is the identified set with each arm's assumption labeled, and the labeling is the substance. An interval reported without its arms is a confession of uncertainty; an interval reported with them is a statement about which assumption would have to be true for which endpoint to hold, and it tells a reader who disagrees with one of the assumptions exactly which number to take. Proposition 3(b) is the model case, because it establishes that a particular decomposition is unavailable in exact terms under the model's normality assumption, and only conditionally recoverable outside it no weakening of the assumption that a reader is likely to grant moves that boundary far, and reporting a point inside it would represent as measured what is in fact chosen.

Fifth and most general, the properties an estimate must satisfy should be asserted in code rather than checked by reading. A long empirical pipeline carries a recurring failure class that no amount of care in the substantive analysis addresses: the wrong-object substitution, in which an estimate of a different quantity, a different population, or a different window flows into a slot whose label it matches and is then reported, cited, and built upon. What defeats it is not vigilance but assertion—automated gates on properties the wrong object cannot satisfy. A truncation bound the quantity must respect. A magnitude scale it must lie on. A sample size, a window length, a standard error inconsistent with the source the value claims to come from. Such gates are cheap to write and they fail loudly, which is the only property that matters, because the wrong object's distinguishing feature is that it is plausible. Taken together the five are a single standard: every number reported about an effect should be accompanied by what the same machinery returns when there is no effect to find, and the accompaniment should be enforced rather than remembered.

Table 2: The correction path, with zero-effect benchmarks

	Switching pairs			Zero-effect benchmark		
Definition	Mean	SD	n	Mean	SD	n
A: mean of log ratios	−0.2837 (0.0257)	1.6630	4,182	−0.6880 (0.0085)	1.1187	17,200
B: log of ratio of sums	0.0758 (0.0244)	1.5799	4,182	−0.2072 (0.0063)	0.8244	17,200
D: B less cell drift	0.2473 (0.0241)	1.5614	4182	—	—	—

Memo. Held-out corrected placebo under D: 0.0188 (0.0087), $n = 8{,}545$.
Split-half reliability under A: 0.9243 ($n = 4{,}120$) against 0.7463 ($n = 15{,}683$).

Notes. Each row is a pair-level effect definition applied to the canonical population of 4,182 switching pairs, and each is paired with the same definition applied to the 17,200 never-treated pairs carrying pseudo-adoption dates, for which the effect is zero by construction. Definition A averages the log ratio of realized to imputed trade over a pair's post-adoption years. Definition B takes the log of the ratio of the sums over the same years, aggregating before the logarithm rather than after, which removes the concavity penalty of Proposition 1. Definition D subtracts from B the mean placebo effect of the pair's size cell, estimated on held-out placebo splits and applied out of sample, which removes the window-geometry drift of Proposition 2. All six treated cells and all four placebo cells come from a single producer on a single pair of populations, so the counterfactual, the exclusion band and the post-window definition are common to every cell shown. The D row has no placebo entry: the correction is defined by subtracting a placebo mean, so its zero-effect benchmark is not a column of this table but the held-out validation of Table [X-REF: T9], where the corrected placebo effect on 8,545 pairs the correction never saw is 0.0188 with a standard error of 0.0087. Reliability is reported separately, in Table [X-REF: T22], because it is computed on the subsets with at least two post-adoption years in each half (4,120 treated, 15,683 placebo) and so does not share this table's row denominators; it is reported for treated and placebo as a pair, since the treated figure alone is uninterpretable—0.9243 against 0.7463 under Definition A. Standard errors in parentheses.

5 What Survives

Table 3 collects what the corrected distribution identifies. Across the 4182 pairs of the canonical population, the agreement effect has a mean of 0.2473 with a standard error of 0.0241 — a positive average effect, precisely estimated, and roughly a fifth of the dispersion around it. The quantity is a log change, so the level-scale counterpart of the average pair is not the exponential of this number: with dispersion of the order documented below, $\mathbb{E}[\exp(\theta)]$ and $\exp(\mathbb{E}[\theta])$ differ materially, and the two should not be substituted for one another. We report effects on the log scale throughout and defer the level-scale translation, where it is needed, to Section 5's successor.

Weighting pairs by the volume of trade they carry rather than counting them equally drives the average down to 0.0898. The weights are pre-adoption trade, not contemporaneous or post-adoption trade, and the choice is not innocuous: post-treatment volume is itself a function of the effect being estimated, so weighting by it would let large estimated effects earn themselves large weights. On pre-adoption weights, the dollar-weighted average agreement effect is roughly a third of the pair-weighted one. That gap is not measurement error. It is the first appearance of a structural feature that organises the rest of this section: the effect is not the same size everywhere, and the places where trade is concentrated are not the places where the effect is large.

The dispersion of true effects is identified only up to an interval. Subtracting the noise-only null from the observed variance leaves a standard deviation of 1.475; subtracting the hardened out-of-sample drift null leaves 0.740; the identified set is [0.74, 1.48], and the two endpoints differ in exactly one assumption — whether the persistent, pair-specific component that the placebo machinery reproduces is counted as noise to be removed or as effect heterogeneity to be retained. The construction is derived in Section 4.1 and we do not prefer an arm here. Both endpoints are large relative to the mean. Under either, the standard deviation of the effect exceeds the average effect by a factor of three or more, and any statement about "the" agreement effect is a statement about the first moment of a distribution whose second moment dwarfs it.

The probability that a randomly chosen pair's true effect is non-positive lies in [0.370, 0.421], where the lower endpoint is the normal-form probability evaluated at the hardened arm and the upper endpoint is capped at the empirical share of non-positive estimates rather than taken from the normal form, which at the noise-only arm returns a value exceeding that share. Both assumptions are load-bearing and neither is innocent: the first imposes a shape the data do not identify, the second substitutes an observed frequency for a modelled one. Separately, 42.11 percent of the pair-level estimates are themselves non-positive, but that figure is a property of the noisy estimates, not of the effects, and it bounds nothing but itself.

What cannot be reported is the shape. At the signal share these data achieve —

roughly 0.37 of the observed variance attributable to true effect variation — the mixture deconvolution does not resolve: fitting three components returns duplicates sitting at the standard-deviation floor rather than distinct subpopulations. This is a limit of the measurement, and it forecloses a family of claims that the literature makes casually. We cannot report the share of pairs that are winners, because that requires locating mass on one side of a threshold in a density we do not have. We cannot report tail asymmetry, or skewness, or the weight of an upper component, or a bulk-and-winners decomposition, because each of those is a statement about shape. We cannot report a point probability of a non-positive effect, which is why the previous paragraph reports a bracket assembled from two assumptions rather than a number.

What survives the limit is more than it might appear. The first moment is point-identified and precisely estimated. The second is set-identified, with both endpoints reported and the assumption separating them named. And the systematic variation in the first moment across observable pair characteristics — the subject of the next subsection — is point-identified throughout, because it requires only conditional means, not a density. A distribution can be unknown in shape while its location, its spread to within an interval, and its dependence on covariates are all recoverable. That is the position these data put us in, and the rest of this section works within it rather than around it.

Table 3: The corrected effect distribution

Quantity	Value
Pairs	4182
Mean effect $\bar{\theta}_D$	0.2473
standard error	(0.0241)
Trade-weighted mean, pre-adoption weights	0.0898
Standard deviation of estimates	1.5614
Standard deviation of true effects, identified set	[0.74, 1.48]
hardened drift arm	0.74
noise-only arm	1.48
$\Pr(\theta \leq 0)$, bracket	[0.370, 0.421]
Share of estimates ≤ 0	0.4211
Signal share	0.37

5.1 The Size Gradient

Sorting pairs into quintiles of pre-adoption trade volume produces the sharpest structural fact in the data. The profile is reported in Table 4. It runs from 0.8554 in the smallest quintile to -0.0583 in the largest, a spread of 0.9137 with a standard error of 0.0809 — eleven standard errors from zero. Two features of its shape carry the argument. The decline is steep across the first three quintiles and then stops: the fourth and fifth quintiles differ by -0.008 against a standard error of 0.046, which is to say they cannot be told apart. And the floor they share is small but not zero, the largest quintile differing from zero at the five percent level where the fourth does not.

The shape matters as much as the slope. Section 2.1 predicts effects that decline in prior integration and then settle at a common floor once the constraints an agreement removes have stopped binding, and it is explicit that a smoothly monotone decline continuing through the largest pairs would count against the mechanism rather than for it: there is no reason for integration to keep reducing an effect that has already been exhausted. The observed profile is a fall and then a floor. The floor is small and slightly negative rather than exactly zero, which the mechanism does not forbid — what it forbids is continued decline — and the two quintiles that constitute it cannot be told apart. This is a more specific prediction than a negative slope, and it is the one the data deliver.

The economic content is straightforward. An agreement removes a set of constraints, and the gain from removing them scales with how tightly they bind. Pairs in the smallest quintile trade an average of well under a million dollars before adoption; pairs in the largest average more than fifteen billion (Table 4). Four orders of magnitude separate them, and they are not the same economic object. For a pair already trading heavily, the frictions an agreement addresses are, by revealed behaviour, not the frictions that were preventing trade; whatever was binding has already been worked around, absorbed into distribution networks, or negotiated away bilaterally. For a pair trading almost nothing, the same removed constraint can be the difference between a market that exists and one that does not. The gradient is what graded effects look like when the grading variable is prior integration.

The gradient is not manufactured by the correction. Computing the same quintile profile on the uncorrected pair-level estimates — before the placebo-based bias correction is applied, reported alongside the corrected profile in Table 4 — returns a smallest-to-largest spread of 0.683 with a standard error of 0.082, which is 74.7 percent of the corrected spread. Three quarters of the gradient is present in the raw estimates. The correction sharpens it and shifts its level, as it shifts every level in this paper, but it does not create it. A reader who distrusts the correction machinery entirely still faces a size gradient of roughly seven tenths of a log point across the trade distribution.

5.2 The Specification Spread, Resolved

The puzzle set out in Section 1 can now be answered. Four specifications of the same gravity equation, on the same data, estimating the same nominal parameter, return agreement effects of 1.402, 0.922, 0.411 and 0.095. The spread is a factor of fifteen. Each of the four is defensible by some standard in the literature, and the discipline has largely

Table 4: Agreement effects by pre-adoption trade quintile

	Corrected (D)		Uncorrected (B)		Pre-adoption
Quintile	Mean	(SE)	Mean	(SE)	trade ($m)
1 (smallest)	0.855	(0.076)	0.557	(0.077)	0.7
2	0.374	(0.060)	0.137	(0.062)	7.8
3	0.131	(0.049)	−0.022	(0.050)	47.1
4	−0.066	(0.037)	−0.168	(0.038)	251.6
5 (largest)	−0.058	(0.028)	−0.126	(0.028)	15,691.0
Spread, 1 − 5	0.914	(0.081)	0.683	(0.082)	

treated the range as a robustness exercise — a band within which the truth presumably lies, to be narrowed by better controls.

Two of the four can be set aside quickly. The first and second omit multilateral resistance terms, and their deviation from the remainder is bias of a kind the literature settled decades ago; nothing in this paper is required to explain why they are wrong.

The genuine puzzle is the third against the fourth: 0.411 from a country-year fixed-effects specification and 0.095 once pair fixed effects are added. Both handle multilateral resistance. Both are standard. The second is the more conservative and is frequently described as such, on the reasoning that identifying from within-pair variation strips out whatever is fixed about a pair and leaves the agreement's own contribution. On that reasoning the smaller number is the more credible one, and the discipline has broadly accepted the ordering.

The results above supply a different reading. If the effect were a constant, every consistent estimator would recover it and the spread would be sampling noise. The effect is not a constant; it is a distribution with a strong slope in pair size. Every estimator therefore reports an average over that distribution, and which average it reports is determined by its implicit weights. A specification whose identifying variation is dominated by large trade flows samples the flat end of the gradient, where effects are near the floor. A pair-fixed-effects specification identifies only from switchers, on a subpopulation defined by the exclusion rules set out in Section 3.3, and weights them non-convexly. Neither is estimating the other's parameter incorrectly. They are estimating different weighted averages of the same sloped object, and the ordering between them carries no presumption about credibility.

The paper's own estimates trace the same pattern. The dollar-weighted mean is 0.0898, close to the fourth specification's 0.095; the unweighted pair-level mean is 0.2473; the country-year specification returns 0.411. Moving from counting pairs to counting dollars moves the average by roughly the same distance as moving between the two defensible specifications. We state plainly what this does and does not establish. It establishes that weighting alone generates spread of the observed magnitude on a distribution with this slope. It does not establish that weighting is the whole of the wedge between the pair-level means and the cross-sectional coefficient: that wedge also contains selection on match quality, since the pairs that sign agreements are not drawn at random from the pairs that could, and this design does not separately identify the two channels. What can be said is the weaker and sufficient claim. The coefficient any one specification reports is an average over a slope, addressed to no particular pair.

5.3 What This Confirms and What It Leaves Open

Section 2.2 advanced four predictions and asked that they be held jointly. Two now carry evidence, and the reason the third does not is worth stating rather than eliding. Lognormal-type outcome tails were to be tested in the distributional diagnostics; no such diagnostic is reported, and the prediction is left untested. What Section 4 does test is the narrower distributional assumption the propositions themselves maintain — that the within-pair log gaps are normal — and that assumption is rejected, with the departure running to the left rather than the right. The mechanism's tail prediction therefore stands neither confirmed nor refuted, and the reader should discount the conjunction accordingly. Effects graded in size and prior integration are confirmed here, and confirmed in the specific fall-then-floor form the mechanism requires rather than merely as a negative slope.

Selection on match quality is supported rather than confirmed, on the placebo and stability evidence assembled in Section 4.3: the pairs that adopt are systematically unlike the pairs that do not, in ways that persist and that the correction layer detects but does not fully purge. Right-skewed effects are neither confirmed nor refuted. The prediction is under-identified at the signal share these data achieve, for the reason given above, and we decline to report the skewness statistics that would appear to settle it. This is the one point in the paper where a prediction that could have been tested is not, and it should be read as a measurement limit rather than as a null result: nothing here weighs against right-skewed effects, and nothing here weighs for them.

The abductive argument therefore stands on three confirmed legs and one open one. That is weaker than the argument as originally posed and stronger than any of its parts taken alone, since the conjunction of lognormal outcomes, graded effects and selection on match quality is not jointly delivered by the alternatives currently on the table. The open leg is recoverable in principle: a longer post-adoption panel, or a research design achieving a higher signal share, would identify the shape and settle the fourth prediction in either direction. It is not recoverable from these data, and we do not claim it.

6 The Boundary of Learnability

6.1 The Equivalence Theorem

The sharpest limit these data impose is not a matter of statistical power, and no larger sample relieves it. Within the post-adoption window, transitory pair-level variation in the effect and a treatment-coincident change in the volatility of trade itself enter the estimating problem only through their sum. Proposition 3(b) states this exactly: conditional on the pair, the post-adoption log gaps are independent and identically distributed as $N(\theta_{ij},\, \sigma_{ij}^2\rho^2 + \omega^2)$, so transitory effect variation and the post-to-pre volatility ratio reach the likelihood only through the composite, and any two parameterizations sharing that composite generate identical data distributions. Distributions that are identical cannot be separated by any statistic computed from them.

The content of that statement is easiest to see as a pair of economies. In the first, the bite of an agreement flickers within the pair: its persistent component is fixed, and a transitory increment of variance is drawn afresh each post-adoption year. In the second, the bite is perfectly steady and what treatment changes is the environment, scaling each pair's pre-adoption dispersion by a common factor. These are different economic claims with different content. Choose the two parameters so that the composites agree, and no statistic distinguishes the economies.

The consequence is the thesis of this section. Under the model's normality assumption, "heterogeneous effects" and "noisier outcomes" are labels for the same data, and the choice between them is a modeling commitment rather than an estimate. Two qualifications follow immediately, and both are material. The complement deserves equal emphasis: Proposition 3(a) identifies the persistent component, the interval this paper reports is an interval in that quantity, and its width is generated by the noise-attribution arms rather than by the equivalence of part (b). And the equivalence is exact only under normality. Outside it, the composite no longer exhausts the likelihood, and Remark C reports what a pooled higher-order moment recovers at this sample: separation of 2.96 to 11.44 under cross-pair independence, against a measured departure that falls inside the simulated range. The boundary is therefore measured rather than absolute, and its position depends on two conditions — cross-pair independence and a shock shape treatment does not alter — neither of which this paper establishes.

6.2 A Warning for Practice

Corollary 2 generalizes what Section 4.2 demonstrated on a single estimator. Define the operational dispersion ratio as a pair's post-window variance over its pre-window variance. Under the model of Proposition 3 that ratio carries the transitory effect variation inside it, indistinguishable from a volatility change, exactly as part (b) requires. A null calibrated by scaling a pre-based noise variance of shorter window by that ratio therefore subtracts the effect variation along with the noise, and over-subtracts by a derivable window factor; at this paper's calibration the whole of the estimated heterogeneity is absorbed at $\kappa_{\text{floor}} = 9.336$. Two properties of this class of estimator are worth stating plainly. The bias is signed and computable rather than a matter of luck. And it is largest where the quantity of interest is largest, since the term inflating the operational ratio is the transitory variance itself: an estimator of this class returns its most reassuring number — heterogeneity near zero — in precisely those worlds where transitory heterogeneity is greatest.

6.3 Bounded Tolerances

Two further limits are bounds rather than walls, and each carries its number. The first is the resolution of the drift calibration. The correction is estimated on held-out placebo pairs and validated on them decile by decile against a bound of 0.10 log points; nine of the ten size deciles clear it, and clear it with room, the largest absolute mean among those nine being 0.0569. The third decile returns −0.1013 on 314 validation pairs and registers PARTIAL. That bound is the resolution of the correction layer, and it governs what may be claimed about fine structure: variation in the size profile smaller than about a tenth of a log point lies inside the calibration tolerance, and this paper makes no claim at that resolution. The fall-then-floor reading of Section 5.1 does not require any. It turns on the collapse from the smallest quintile to the floor, which exceeds the tolerance by nearly an order of magnitude, and it asserts nothing about the ordering of the quintiles constituting the floor.

The second is the distributional assumption itself. The propositions are derived under normality of the within-pair log gaps, and that assumption is tested directly in Section 4 and rejected, at 32.8 percent of pairs against 5.0 percent for a matched control. Two things follow, and they pull in opposite directions. The predictions the propositions generate survive the failure of their own premise — the reliability formula lands within eight thousandths of the observed value on the same data — which is evidence about the mechanism rather than about the premise. But the equivalence of Section 6.1 was exact only under that premise, and its failure is what opens the higher-moment route. The paper reports both rather than choosing.

6.4 The Transportability Boundary

The estimand is a distribution of effects over self-selected matches, and that is a limit of the same standing as the equivalence theorem: proven by the design rather than conjectured about it. The estimating population consists of pairs that adopted an agreement, and the counterfactual against which their effects are measured is fitted on untreated cells only (Section 3.4). Nothing in that construction identifies the effect of an agreement assigned to a pair that did not choose one. No such pair appears in the estimating population, and no assumption available to the design manufactures one. This is stronger than the customary caution about external validity, which concerns whether an estimate travels: here the population over which the estimand is defined does not contain the pairs to which a reader might wish to extrapolate.

The margin at issue is the one the theory names in advance. Section 2.3 predicts that agreements are selected on the same unobserved conditions that modulate their effects, and the placebo and stability evidence of Section 4.3 supports that prediction. The prediction and the limit are one fact seen from two sides: the selection that makes the mechanism testable is the selection the estimand conditions on. That conditioning is the constraint any policy reading must respect.

The identification ledger

Table 5: The identification ledger

Quantity	Status	Instrument
Dispersion of persistent effects	Set-identified	Prop. 3(a); arm accounting
Transitory/volatility split	Conditionally unidentified	Prop. 3(b); Rem. C
Drift-calibration residual	Bounded at 0.10	Out-of-sample decile validation
Effect on a non-adopting pair	Unidentified by design	Estimating-population construction

Each row records a quantity this design bears on, its status, and the instrument establishing that status. "Set-identified" means an interval is recovered whose endpoints differ in an assumption the data do not adjudicate. "Conditionally unidentified" means exactly unidentified under the model's normality assumption and recoverable outside it only under conditions this paper does not establish. "Bounded" means measured up to a tolerance a gate on held-out data certifies. No row reports a status stronger than its instrument supports.

None of these four limits is assumed. The first is a theorem, the second a corollary of it, the third the reported outcome of a gate run on held-out data, and the fourth a property of a population whose construction is tabulated. Each also names its own exit. The equivalence is broken by exogenous variation in outcome volatility, or by non-normality sufficient to make pooled higher-order moments informative — a route this paper measures but does not walk. The over-subtracting null is defeated by matching the null's window to the estimation window, which is a design choice rather than a discovery. The calibration tolerance narrows with cells that are finer and still large enough to certify. The transportability boundary is crossed only by variation in assignment. What remains between the limits is the shape of this paper's contribution: the diagnostics that could be computed were confidently wrong, the quantities that mattered most are bounded rather than free, and the section between those two facts is where this paper's positive claims live.

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A Additional Tables and Figures

B Proofs

C Propositions

Notation and assumptions

y_{ijt} observed trade; $\hat{y}_{ijt}$ the PPML-imputed counterfactual; $\eta_{ijt} := y_{ijt}/\hat{y}_{ijt}$; $g_{ijt} := \log \eta_{ijt}$. Definitions: $\theta^A_{ij} := T^{-1} \sum_{t\in\text{post}} g_{ijt}$; $\theta^B_{ij} := \log\big(\sum_t y_{ijt}/\sum_t \hat{y}_{ijt}\big)$ over the post window.

Assumption 1 (A1). *$y_{ijt} = \hat{y}_{ijt}\eta_{ijt}$ with $\mathbb{E}[\eta_{ijt}] = 1$.*

Assumption 2 (A2). *$\log \eta_{ijt} \sim N(-\sigma^2_{ij}/2, \sigma^2_{ij})$, i.i.d. over t within pair.*

Assumption 3 (A3). *σ^2_{ij} varies across pairs and is a persistent pair trait.*

Proposition 1 (Jensen bias and its reliability). *Under (A1)–(A2) with zero treatment effect:* (a) *$\mathbb{E}[\theta^A_{ij} \mid ij] = -\sigma^2_{ij}/2$ exactly.* (b) *Write $R_{ij} = \sum_t w_t\eta_{ijt}$ with $w_t = \hat{y}_{ijt}/\sum_s \hat{y}_{ijs}$. Then $\mathbb{E}[\theta^B_{ij}] = -\frac{1}{2}\operatorname{Var}(R_{ij}) + r_{ij}$, where $\operatorname{Var}(R_{ij}) = \operatorname{Var}(\eta_{ij})\sum_t w_t^2$ (equal weights: $\operatorname{Var}(\eta_{ij})/T$), $\operatorname{Var}(\eta_{ij}) = e^{\sigma^2_{ij}} - 1$, and the remainder r_{ij} collects terms of order $\mathbb{E}[(R-1)^3]$ and higher; the expansion requires $\operatorname{Var}(R_{ij}) \ll 1$.* (c) *For split-half estimates on disjoint post halves of length T_h,*

$$\operatorname{Corr}\big(\theta^{A,(1)}, \theta^{A,(2)}\big) = \frac{\operatorname{Var}_{pairs}(\sigma^2_{ij})/4}{\operatorname{Var}_{pairs}(\sigma^2_{ij})/4 + \mathbb{E}[\sigma^2_{ij}]/T_h},$$

exactly under (A2).

Proof. (a) Immediate from (A2). (b) Second-order expansion of $\log R$ about $R = 1$ with $\mathbb{E}[R] = 1$; the weight identity and $\operatorname{Var}(\eta) = e^{\sigma^2} - 1$ (from $\mathbb{E}[\eta^2] = e^{\sigma^2}$ under (A2)) give the display. (c) Conditional on the pair, $\theta^{A,(h)} = -\sigma^2_{ij}/2 + \bar{\varepsilon}^{(h)}$ with $\bar{\varepsilon}^{(h)} \sim (0, \sigma^2_{ij}/T_h)$ independent across halves; decompose variance and covariance across pairs. □

Corollary 1 (Reliability paradox). *Let b_{ij} be* any *persistent per-pair bias with cross-pair variance* $\operatorname{Var}(b)$*; then under zero effects $r_{\text{split}} = \operatorname{Var}(b)/\big(\operatorname{Var}(b) + \mathbb{E}[\sigma^2]/T_h\big)$. High split-half reliability certifies persistence, not validity: any dispersed, persistent artifact (Jensen, drift) passes.*

Remark (Empirical scale; validity of the expansions). *Part (a) identifies the mean dispersion directly from the zero-effect population, since $\mathbb{E}[\theta^A] = -\mathbb{E}[\sigma^2]/2$ exactly: at the R-chain placebo mean -0.6821 under Definition A this gives $\mathbb{E}[\sigma^2] = 1.3642$ and $\sigma = 1.1680$. At that dispersion $\operatorname{Var}(\eta) = 2.9126$, so the generic Taylor form $-\operatorname{Var}(\eta)/2$ is* invalid *for θ^A and the exact lognormal value applies; and for θ^B at $T = 10$, $\operatorname{Var}(R) = 0.2913$, so $-\operatorname{Var}(R)/2 = -0.1456$ is a rough guide only. Simulation at the same (σ, T) gives*

$\mathbb{E}[\theta^B] = -0.112$*: the second-order expansion overstates the bias magnitude by about* 30 *percent, the positive third-order (skewness) term partially offsetting the variance term, so the approximation is conservative at this scale. The R-chain B-placebo mean* -0.2072 *additionally contains drift (Proposition 2). Consistency check [V1c]: part (c) is over-identified here, and the check is parameter-free. The cross-pair variance of* θ^A *on the zero-effect population decomposes as* $\mathrm{Var}(\sigma^2)/4 + \mathbb{E}[\sigma_{ij}^2/T_{ij}^{\mathrm{post}}]$*, and the second term must be evaluated pair by pair rather than at the mean window length: post windows differ across pairs,* $1/T$ *is convex, and the reciprocal of the mean understates the mean of the reciprocals. Evaluated pair-wise, the decomposition yields* $\mathrm{Var}(\sigma^2) = 4.0210$*, a coefficient of variation of* σ_{ij}^2 *of* 1.470*. Substituting that into the split-half formula, again pair-wise,* predicts *a placebo reliability of* 0.7539 *against the observed* 0.7463 *— a discrepancy of* 0.0076*, over-predicting slightly, with no parameter fitted to it. Evaluating the same decomposition at the mean window length instead returns* $\mathrm{Var}(\sigma^2) = 4.1773$ *and a reliability whose error is roughly eight times larger, in the direction convexity requires. The pair-wise evaluation is therefore not a refinement of the proposition but the correct reading of it.*

Proposition 2 (Within-pair drift). *Relax (A2) to* $\log \eta_{ijt} = -\sigma_{ij}^2/2 + \delta_{ij}\,(t - \bar{t}_{ij}) + u_{ijt}$*,* $u_{ijt} \overset{iid}{\sim} N(0, \sigma_{ij}^2)$*, with* δ_{ij} *a persistent pair-specific drift and* $\bar{t}_{ij}$ *the pair's sample midpoint (the pair fixed effect centers the level, not the trend). For a never-treated pair with span* $T = T_{\mathrm{pre}} + T_{\mathrm{post}}$ *and post window equal to the last* T_{post} *years:* (a) $\overline{(t-\bar{t})}^{\,\mathrm{post}} = T_{\mathrm{pre}}/2$*, hence to first order* $\mathbb{E}[\theta_{ij}^A] = -\sigma_{ij}^2/2 + \delta_{ij} T_{\mathrm{pre}}/2$ *(and* θ^B *inherits the same drift term with* $\hat{y}$*-weighted window mean); the expression depends on window geometry only, never on calendar time.* (b) *The drift term is a persistent pair trait and passes split-half reliability by Corollary 1.* (c) *Now let true effects* $\theta_{ij} \sim (\mu, \tau^2)$ *enter the post window. If* δ_{ij} *correlates with pair size* s*, the Definition-D correction (subtracting the size-cell placebo mean) removes* $\mathbb{E}[\delta_{ij} \mid s]\, T_{\mathrm{pre}}/2$ *but not the within-cell dispersion:*

$$\mathrm{Var}(\theta^D \mid s) \;=\; \tau^2 + \big(T_{\mathrm{pre}}/2\big)^2\, \mathrm{Var}(\delta_{ij} \mid s) + \frac{\mathbb{E}[\sigma_{ij}^2 \mid s]}{T_{\mathrm{post}}},$$

so any deconvolution recovers τ^2 *jointly with the within-cell drift variance unless the null design carries matching window geometry.*

Proof. (a) The post-window mean of t is $T_{\mathrm{pre}} + (T_{\mathrm{post}}+1)/2$; subtracting $\bar{t} = (T+1)/2$ gives $T_{\mathrm{pre}}/2$. Calendar year enters nowhere. (b) Apply Corollary 1 with $b_{ij} = -\sigma_{ij}^2/2 + \delta_{ij} T_{\mathrm{pre}}/2$. (c) Decompose $\theta^D = \theta_{ij} + \big(\delta_{ij} - \mathbb{E}[\delta \mid s]\big) T_{\mathrm{pre}}/2 + \bar{u}$, take variances conditional on s. □

Remark. *Part (a) predicts a placebo bias governed by window geometry alone and, where* δ_{ij} *correlates with size, graded in size. Simulation confirms both: at* $\delta = 0.02$ *the mean pseudo-effect matches the predicted* -0.5821 *at* -0.5808*; shifting every pair's span by an* *arbitrary calendar offset leaves the result unchanged (shift* ≈ 0.0010*, confirming calendar invariance). Empirically the R-chain placebo means under Definition B are graded over the upper size deciles, falling in magnitude from* -0.365 *at the fifth decile to* -0.057 *at the largest, and are non-monotone below the fifth: the smallest decile returns* -0.206 *against an overall mean of* -0.207*. The proposition predicts grading conditional on the size–drift correlation and does not predict monotonicity across the full support; the observed profile is reported as it stands.*

Proposition 3 (Identification and its boundary)**.** *Let pre-window gaps be* $g_{ijt} = \sigma_{ij} u_{ijt}$ *(mean-adjusted) and post-window gaps* $g_{ijt} = \theta_{ij} + \nu_{ijt} + \sigma_{ij}\rho\, u_{ijt}$*, with* $u_{ijt} \overset{iid}{\sim} N(0,1)$*, transitory effect variation* $\nu_{ijt} \overset{iid}{\sim} N(0,\omega^2)$ *independent of* u*, persistent effects* $\theta_{ij} \sim (\mu, \tau^2)$*, and post/pre volatility ratio* $\rho \geq 0$*. Define the per-pair observables: pre variance* V_{ij}^{pre}*, centered post variance* V_{ij}^{post}*, and* $\hat{\theta}_{ij} = \bar{g}^{\text{post}}$*. Then:* (a) (Identification of τ^2.) $\mathbb{E}[V_{ij}^{\text{post}}] = \sigma_{ij}^2\rho^2 + \omega^2$, $\text{Var}(\hat{\theta}_{ij} \mid ij) = V_{ij}^{\text{post}}/T_{\text{post}}$ *exactly, hence*

$$\tau^2 = \text{Var}_{pairs}(\hat{\theta}) - \mathbb{E}\big[V^{\text{post}}\big]/T_{\text{post}},$$

with all right-hand quantities estimable (the within-pair sample variance is unbiased for V^{post}*).* (b) (Exact non-identification of (ω^2, ρ).) *Under the stated normality, the post gaps are, conditionally on the pair, i.i.d.* $N(\theta_{ij},\, \sigma_{ij}^2\rho^2 + \omega^2)$*: the parameters enter the likelihood only through the composite* $\sigma^2\rho^2 + \omega^2$*. Any two parameterizations with equal composites generate* identical *data distributions; the split between transitory effect heterogeneity and treatment-coincident volatility change is a modeling label, not an estimable quantity. The equivalence is exact under the stated normality, and is claimed at that strength and no further. Absent normality the composite no longer exhausts the likelihood, and separation becomes possible in principle through within-pair moments of order* ≥ 3*; the sampling noise of any* per-pair *such moment overwhelms it at* $T_{\text{post}} \approx 10$*, but the operative statistic pools across pairs and its noise falls with the number of pairs, not the window length. Remark C reports what that pooled route delivers at this sample.*

Corollary 2 (Operational variance ratios over-subtract)**.** *Define* $\hat{\rho}_{\text{op}}^2 := V^{\text{post}}/V^{\text{pre}} = \rho^2 + \omega^2/\sigma^2$*. A "post-calibrated" null that scales a* pre-based *noise variance of window length* $T_0 < T_{\text{post}}$ *by* $\hat{\rho}_{\text{op}}^2$ *subtracts* $\kappa_w\, V^{\text{post}}/T_{\text{post}}$ *with window factor* $\kappa_w := T_{\text{post}}/T_0 > 1$*, biasing the recovered variance by* $-(\kappa_w - 1)\,\mathbb{E}[V^{\text{post}}]/T_{\text{post}}$*. At* κ_{floor} *the entire treatment variance is absorbed:* $\hat{\tau}^2(\kappa_{\text{floor}}) = 0$*.*

Remark (The empirical bracket, organized by arm)**.** *Part (a) is precisely the window-matched estimator used in the R-chain V1 computation. The* SD_{true} *bracket* $[0.74, 1.48]$ *(INV-027) is arm-indexed: Arm C (OOS drift subtraction,* $\text{Var}_{\text{null}} = 1.89$*) yields* $\text{SD}_{\text{true}} = 0.74$*; Arm A (noise-only subtraction,* $\text{Var}_{\text{null}} = 0.26$*) yields* $\text{SD}_{\text{true}} = 1.48$*. Under the stated model, the interval spans the admissible noise attributions. Corollary 2 quantifies* *what a mis-specified null window costs. At window factor $\kappa = 1$ the subtraction is the noise-only arm and returns its* $\mathrm{SD}_{\mathrm{true}}$ *of* 1.4754 *by construction; at* $\kappa = 2, 3, 5$ *the recovered dispersion falls to* 1.3841*,* 1.2863 *and* 1.0641*, reaching zero at* $\kappa_{\mathrm{floor}} = 9.336$*. A null calibrated on a window shorter than the estimation window therefore absorbs true heterogeneity at a computable rate, and at* κ_{floor} *absorbs all of it. The transitory slice* ω^2/T *is, by part (b), exactly inseparable from volatility change under normality; Remark C bounds what departures from normality recover.*

Remark (What departures from normality recover)**.** *Part (b)'s equivalence is exact under normality and silent outside it. The question outside it is quantitative: pooled across pairs, does a higher-order within-pair moment separate the two parameterizations at this sample? A calibrated simulation answers it. Drawing* σ^2_{ij} *to match the ledgered first two moments and using the empirical distribution of post-window lengths, the two worlds are constructed with equal composites and compared on the pooled mean of per-pair sample excess kurtosis. Under normality the difference is zero to simulation precision, as part (b) requires. Away from it, and expressed as the ratio of the difference to the sampling standard deviation of the statistic in a* single *realization of* 4,182 *pairs, separation runs from* 2.96 *to* 11.44 *across excess kurtosis in* [0.75, 3.0] *and transitory shares in* [0.10, 1.00]*. The relevant departure is measured rather than assumed: the standardized log gaps carry an excess kurtosis of* 0.784 *against* -0.269 *for a matched normal control passed through identical code, a net departure near unity that falls inside the simulated grid.*

Two qualifications bound the reading, and both are material. The simulation treats pairs as independent; cross-pair dependence is not estimated anywhere in this paper, and at assumed variance inflations of ten and fifty the same separation falls to 3.62 *and* 1.62 *at the most favorable cell. And the route requires committing to a shape for the multiplicative shock that treatment does not alter; if treatment may change the shape as well as the scale, the separation closes again. The honest statement is therefore conditional. The split between transitory effect heterogeneity and treatment-coincident volatility change is exactly unidentified under normality; outside normality it is recoverable at this sample under cross-pair independence and shape invariance, and its status without those two conditions is not established here. Nothing in this paper's reported interval depends on which way that resolves, since the interval concerns the persistent component identified by part (a).*